Early Pleistocene volcanism and glaciation in central Yukon: a new chronology from field studies and paleomagnetism[1]

L.E. Jackson, Jr., R.W. Barendregt, J. Baker, and E. Irving

Abstract: The paleomagnetism of the Selkirk Volcanics and nearby stratified Pleistocene sediments was investigated to resolve the chronology of Early Pleistocene glaciations in central Yukon. Radiometric dates on these low-K basalts have proven to be erroneously old. Most sampled sediments and all basalts accurately record the paleofield and true reversals. The valley-filling phase of the Selkirk Volcanics was in part coeval with the younger pre-Reid glaciation. It was erupted during the Matuyama Chron, either post-Cobb Mountain Subchron or post-Jaramillo Subchron, over a period too brief to average secular variation. The older pre-Reid glaciation occurred after ca. 1.60 Ma and prior to the eruption of the Fort Selkirk tephra (pre-Jaramillo or pre-Cobb Mountain). Sediments investigated at Revenue Creek and Braden's Canyon are normally magnetized. The assigned Brunhes age is compatible with their occurrence in valleys that were cut or deepened sometime after the pre-Reid glaciations.

Résumé : Le paléomagnétisme des Volcanites de Selkirk et des sédiments stratifiés à proximité d'âge Pléistocène a été étudié pour dater les glaciations qui marquèrent le début du Pléistocène dans la partie centrale du Yukon. Nous démontrons que les âges radiométriques disponibles de ces basaltes pauvres en K sont trop anciens. La plupart des sédiments échantillonnés et tous les basaltes ont enregistré avec exactitude le paléochamp et les inversions réelles. Le remplissage des vallées par les Volcanites de Selkirk fut en partie contemporain de la plus jeune avancée glaciaire antérieure à la Glaciation de Reid. Les volcanites ont fait éruption durant le Chron de Matuyama, qui est postérieur au Sous-chron de Cobb Mountain ou au Sous-chron de Jaramillo, elles furent de trop courte durée pour pouvoir estimer une variation séculaire moyenne. La plus encienne avancée glaciaire antérieure à la Glaciation de Reid est apparue ultérieurement à plus ou moins 1,60 Ma, et avant l'éruption anté-Jaramillo ou anté-Cobb Mountain du téphra de Fort Selkirk. Les sédiments étudiés des sites de Revenue Creek et de Braden's Canyon sont caractérisés par une aimantation normale. L'âge de Brunhes qui leur est assigné est compatible avec le fait que ces sédiments sont présents dans les vallées entaillées par l'érosion ou surcreusées, tôt ou tardivement, après les avancées glaciaires qui ont précédé la Glaciation de Reid.
[Traduit par la rédaction]

Introduction

In central Yukon, Quaternary sediments record several glaciations and interglaciations or interstadials (Bostock 1966; Hughes et al. 1969) (Fig. 1). In the Fort Selkirk area, glacial drift from the oldest of the pre-Reid glaciations and tephra-bearing nonglacial sediments are stratified between basalt and basaltic hyaloclastite flow complexes. The overlying basalt was erupted during the younger of the pre-Reid glaciations (Jackson et al. 1990). It would appear that radiometric dating of basalt and tephra alone would allow an accurate chronology of early Cordilleran glaciation to be established. However, K–Ar and fission-track ages have lacked concordancy (Naeser et al. 1982; Westgate 1989).

Up to now, paleomagnetic investigation of the Selkirk Volcanics and interstratified glacial sediments has been limited. Dubois (1959) made the first paleomagnetic investigation of the Selkirk Volcanics and determined that both normal and reversed polarity occurred. Further limited results that corroborated DuBois' findings were reported by Naeser et al. (1982). Jackson et al. (1990) made a preliminary report of an extensive investigation of the magnetostratigraphy of the Selkirk Volcanics and interstratified sediments. They proposed several possible stratigraphic schemes based on the magnetic polarities observed.

This paper reports the results of subsequent work in the application of magnetic polarity to constrain the ages of early Quaternary glaciations in central Yukon using the magnetic polarity time scale.

Regional Quaternary geology setting

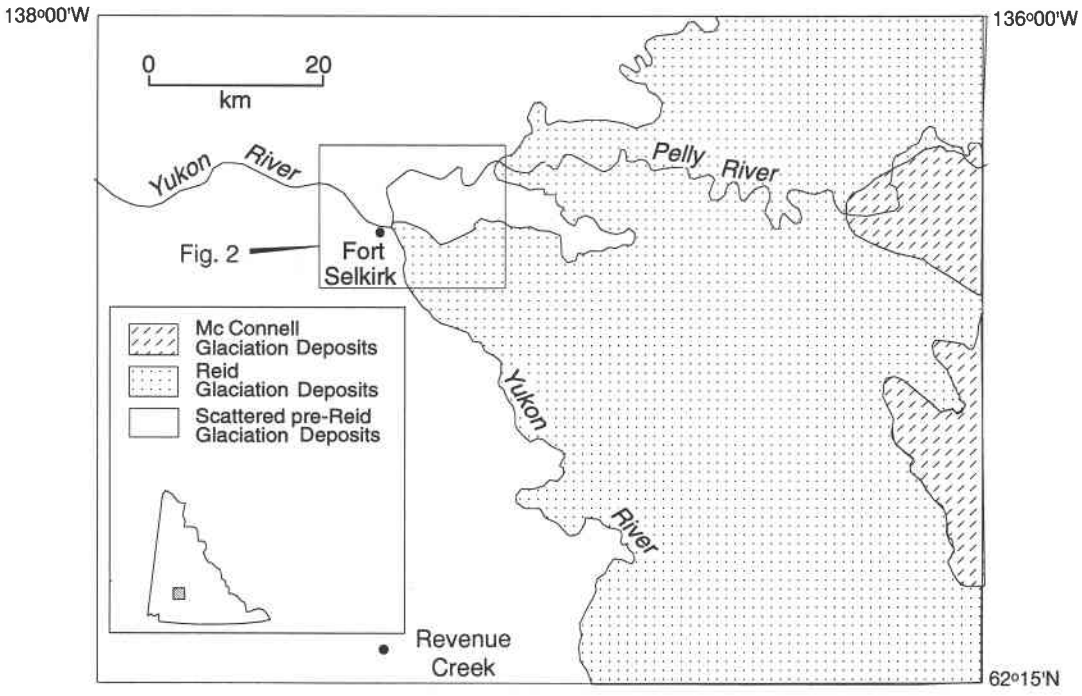
The area of investigation lies within the Yukon Plateau (Mathews 1986), a rolling accordant and incised upland with elevations ranging from 430 m along the Yukon River to 1938 m at the summit of Klaza Mountain in the Dawson

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L.E. Jackson, Jr.[2] Terrain Sciences Division, Geological Survey of Canada, 100 West Pender Street, Vancouver, BC V6B 1R8, Canada.
R.W. Barendregt. Department of Geography, The University of Lethbridge, Lethbridge, AB T1K 3M4, Canada.
J. Baker and E. Irving. Pacific Division, Geological Survey of Canada, 9860 West Saanich Road, Sidney, BC V8L 4B2, Canada.

[1] Geological Survey of Canada Contribution 1996144.
[2] Corresponding author (e-mail: ljackson@gsc.emr.ca).

Fig. 1. Locality map showing Fort Selkirk, Revenue Creek, and glacial limits in central Yukon.



Range. Relief seldom exceeds 600 m. The area has a subarctic climate and permafrost is widespread. Bostock (1966) recognized four glaciations in central Yukon, which he named, from youngest to oldest, McConnell, Reid, Klaza, and Nansen. The deposits and landforms of the latter two are generally not easily distinguished and have been lumped together as pre-Reid (Hughes et al. 1969). Subsequent applications of radiocarbon and K–Ar dating, tephrochronology, paleopedology, and general stratigraphic investigations have shown the McConnell Glaciation to be Late Wisconsinan (Matthews et al. 1990; Jackson and Harington 1991), the Reid Glaciation to predate the last interglacial (Smith et al. 1986; Hughes 1987; Hughes et al. 1989; Westgate 1989), and the pre-Reid glaciations to predate the Brunhes Chron (Naeser et al. 1982; Jackson et al. 1990). The chronology of the pre-Reid glaciations remains the least controlled of the four known glaciations in central Yukon.

The Old Crow tephra overlies Reid-age drift. It has yielded a mean fission-track age of 140 ± 10 ka, providing a minimum age for the Reid Glaciation (Westgate et al. 1990). Basalts underlying Reid drift are reversely magnetized and are referred to the Matuyama Chron (see below). Consequently, the Matuyama–Brunhes boundary (ca. 0.78 Ma; Shackleton et al. 1990) serves both as a minimum limiting age for the pre-Reid glaciations (Jackson et al. 1990) and as a maximum limiting age for the Reid Glaciation.

Basalt flows in excess of 100 m thick are found along the Yukon and Pelly rivers in the area of Fort Selkirk (Dawson 1889; Hayes 1892). They were first described in detail by Bostock (1936), who named them Selkirk Volcanics. Included are the pillow basalts, breccias interstratified with lava flows, and tuff and breccia that make up the volcanic peaks of Volcano Mountain and Ne Ch'e Ddhäwa (aboriginal name) and surrounding complexes of lava flows (Fig. 2) (Jackson

Fig. 2. Sampled and described sections in the Fort Selkirk and Braden's Canyon area. Selkirk Volcanics denoted by the honeycomb pattern.

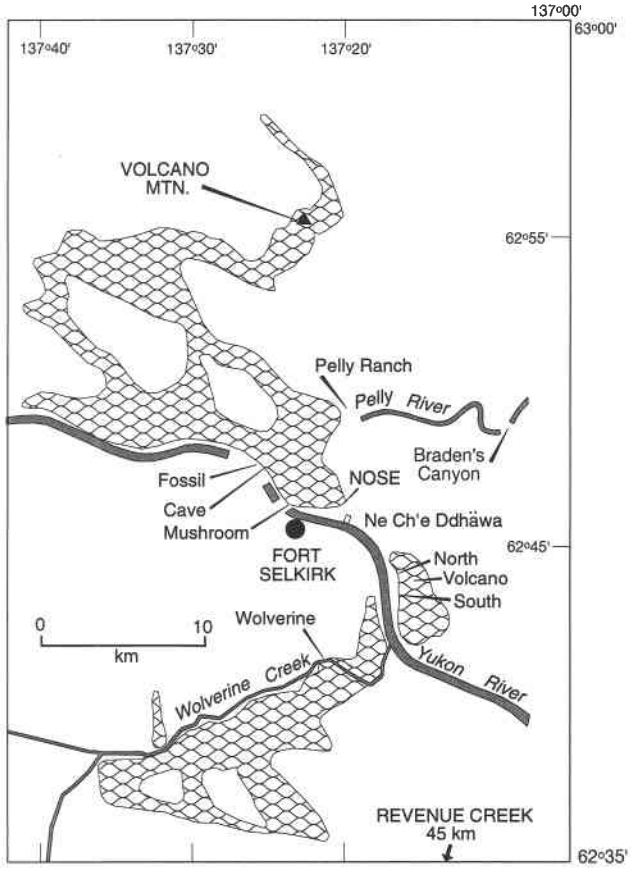
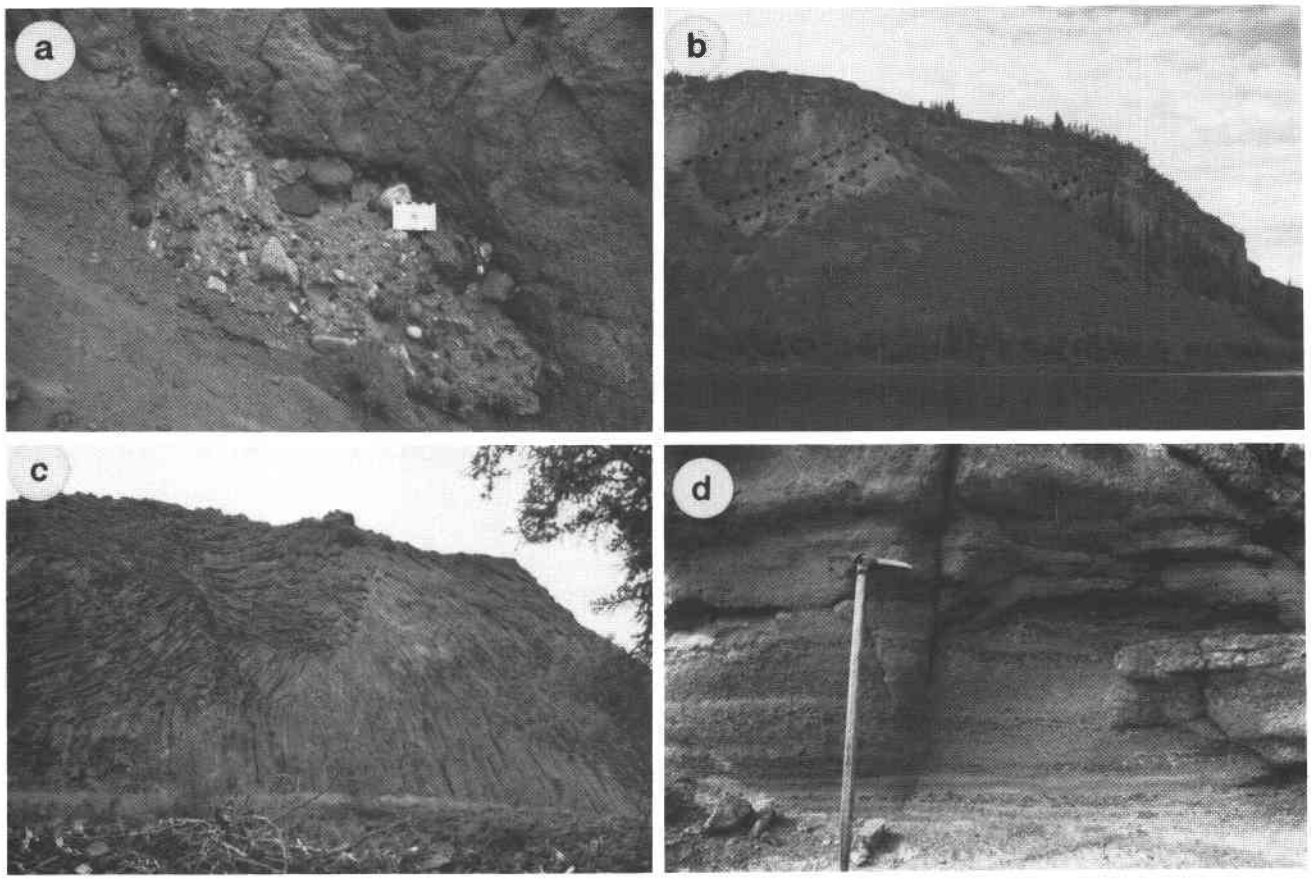


Fig. 3. Example volcanic facies. (*a*) Lens of glacially derived diamicton incorporated within massive tuff breccia. (*b*) Nose section: foreset-bedded pillow basalt (bedding planes are accentuated by dots) capped with a series of colonnade–entablature successions (cliff is 60 m high). (*c*) Radially fanning "war bonnet" columnar basalt (exposure is 40 m high). (*d*) Stratified tuff breccia (pick is 1 m long).



1989; Jackson et al. 1990; Jackson and Stevens 1992). The Selkirk Volcanics commonly contain ultramafic (spinel lherzolite) xenoliths (Sinclair et al. 1978), and exotic pebbles or glacial diamicton in the case of Ne Ch'e Ddhäwa (Fig. 3*a*). The Selkirk Volcanics had at least three eruptive sources: vents along upper Wolverine Creek, Ne Ch'e Ddhäwa, and the Volcano Mountain area (Bostock 1936; Frances and Ludden 1990) (Fig. 2). Their compositions range from nephelinite to basanite and fall within the alkaline olivine basalt field series (Frances and Ludden 1990). The valley-filling basalt and hyaloclastite of the Selkirk Volcanics were erupted during the Early Pleistocene (Owen 1959*a*, 1959*b*; Naeser et al. 1982). However, eruptions of Volcano Mountain could have occurred as recently as the early Holocene (Jackson and Stevens 1992).

This study

The initial investigation of Jackson et al. (1990) provided several possible interpretations of the ages of the pre-Reid glaciations. A key element is the occurrence of normally magnetized fine sand and silt beds in an otherwise reversely magnetized sediment sequence interstratified with Selkirk Volcanics. Nearly continuous natural sections through Selkirk Volcanics north and south of Fort Selkirk and sedimentary sequences at Braden's Canyon 13 km northwest from Fort Selkirk and an artificial exposure in a placer mine at Revenue Creek 50 km south of Fort Selkirk were measured, described, and sampled for texture, sediment provenance, and K–Ar and ^{14}C dating as required during the summers of 1990–1993 (Fig. 2). All the sedimentary sections proved to be beyond the range of radiocarbon dating (Table 1) or contained vertebrate fossil remains of sufficient antiquity to rule out radiocarbon dating (see below), leaving paleomagnetic investigation as the only means of age control.

Table 1. Radiocarbon ages.

Location (see Fig. 4)	Sample No.	Age (years BP)	Depth (m)	Material
Braden's Canyon	GSC-4510	>41 000	−30	Degraded peat
Revenue Creek	GSC-4963	>38 000	−2	Wood
Revenue Creek	GSC-4935	>40 000	−4	Wood

Sedimentary and volcanic facies and environments

The Selkirk Volcanics, excluding Volcano Mountain, were subdivided into eight volcanic facies based on jointing patterns,

Table 2. Volcanic facies and eruptive environments, Fort Selkirk area.

Facies	Thickness (m) Area (m^2)	Association	Architecture and lithology	Eruptive environment
V-1: columnar and blocky jointed basalt	4–5 10^5–10^6	Typically overlies V-3 and V-4	Colonnade–entablature successions; colonnades composed of prismatic basalt columns 30–100 cm in diameter	Vertical jointing and large aerial extent indicate emplacement as subaerial lava flows (Waters 1960)
V-2: "war bonnet" columnar basalt (after Waters 1960)	3–10 10^2–10^4	Contacts V-3 and V-5 laterally and vertically and overlies V-3	Radially fanning, recumbent, and contorted basalt columns, which locally give the impression of feathers arrayed on a war bonnet (Fig. 3*c*)	Fanning and recumbent columns are indicative of subaqueous or subglacial eruption and flow (Mathews 1958; Roddick et al. 1977; Bergh 1985; Bergh and Sigvaldson 1991)
V-3: subflow breccia	<1 10^3–10^4	Underlies V-1 and V-2 and grades vertically downward into V-4	Oxidized and altered angular and vesicular fragments of basalt; fragments range from <1 to 30 cm; pillows and pillow fragments are common	Subflow breccias form from fragmentation of the basal zone of lava flows due to rapid quenching where they traverse damp ground or shallow water
V-4: foreset-bedded pillow breccia and massive pillow breccia	5–65 10^5–10^6	Overlain by V-1, with V-3 locally present as an intervening transition zone	Accumulations of basalt pillows and pillow fragments; these may form foreset beds; pillows commonly have fresh rinds of sideromelane up to 2 cm thick (Fig. 3*b*)	Foreset-bedded pillow breccias form when lava flows advance into deep water, whereas massive pillows are indicative of eruption within water (Russell 1902; Fuller 1931; Jones and Nelson 1970; Moore et al. 1973; Furnes and Fridlefsson 1974; Furnes and Sturt 1976)
V-5: pillow tuff breccia	200 10^5	Intergrades with V-4	Basalt hyaloclastite breccia: a chaotic mixture of angular pillow fragments in a matrix of sandy tuff altered to kaolinite, mixed-layered clays, and zeolite (Fig. 3*a*); stratification commonly deformed by syndepositional slumping	Makes up most of the Ne Ch'e Ddhäwa volcanic edifice; contains exotic pebbles incorporated during subglacial eruption of Ne Ch'e Ddhäwa (Jackson 1989; Hickson 1986, 1990)
V-6: stratified massive tuff and tuff breccia	0–100 10^5	Grades into V-5 toward volcanic centre	Altered, stratified basaltic tuff and lapillite (Fig. 3*d*); original pyroclasts have been altered to kaolinite, mixed-layered clays, zeolite (analcime), albite, and calcite	Fine texture, hyaloclastite lithology, exotic pebbles, and position at the margin of Ne Ch'e Ddhäwa are consistent with a distal subglacial volcanic eruptive environment
V-7m: basaltic lapilli	1–2 10^5	Overlain by V-1 and V-3	Coarse vitreous basaltic ash (<4 mm) coarsening upward to lapilli (4–<32 mm)	Tephra from a local eruptive centre; locally contains carbonized wood at its base
V7f: felsic ash	10–20 cm 10^4?	Interstratified with fine sand	Primary ash bed is commonly succeeded by thin, impure, reworked beds of ash, silt, or fine sand	Tephra from a remote felsic volcanic centre (Naeser et al. 1982)

flow architecture, texture, and sedimentary structures within hyaloclastic and pyroclastic units. These are summarized in Table 2. Examples of some of these facies are presented in Fig. 3, and the stratigraphy of the sections studied is summarized in Fig. 4. The Selkirk Volcanics were emplaced as subaerial flows, flows advancing into or beneath standing water (Fig. 3*b*), subglacially erupted hyaloclastite tuff and pillow tuff breccia (Fig. 3*d*), and tephra.

Sediments interstratified within the Selkirk Volcanics and those at Braden's Canyon and Revenue Creek were subdivided into facies suites using a modified version of the facies notation scheme of Eyles and Miall (1984). Sediments

Fig. 4. Summary of sedimentary and volcanic facies and sample sites for measured and described sections. The strip log to the right of each section represents polarity: shaded area indicates normal magnetization, and unshaded indicates reversed magnetization.

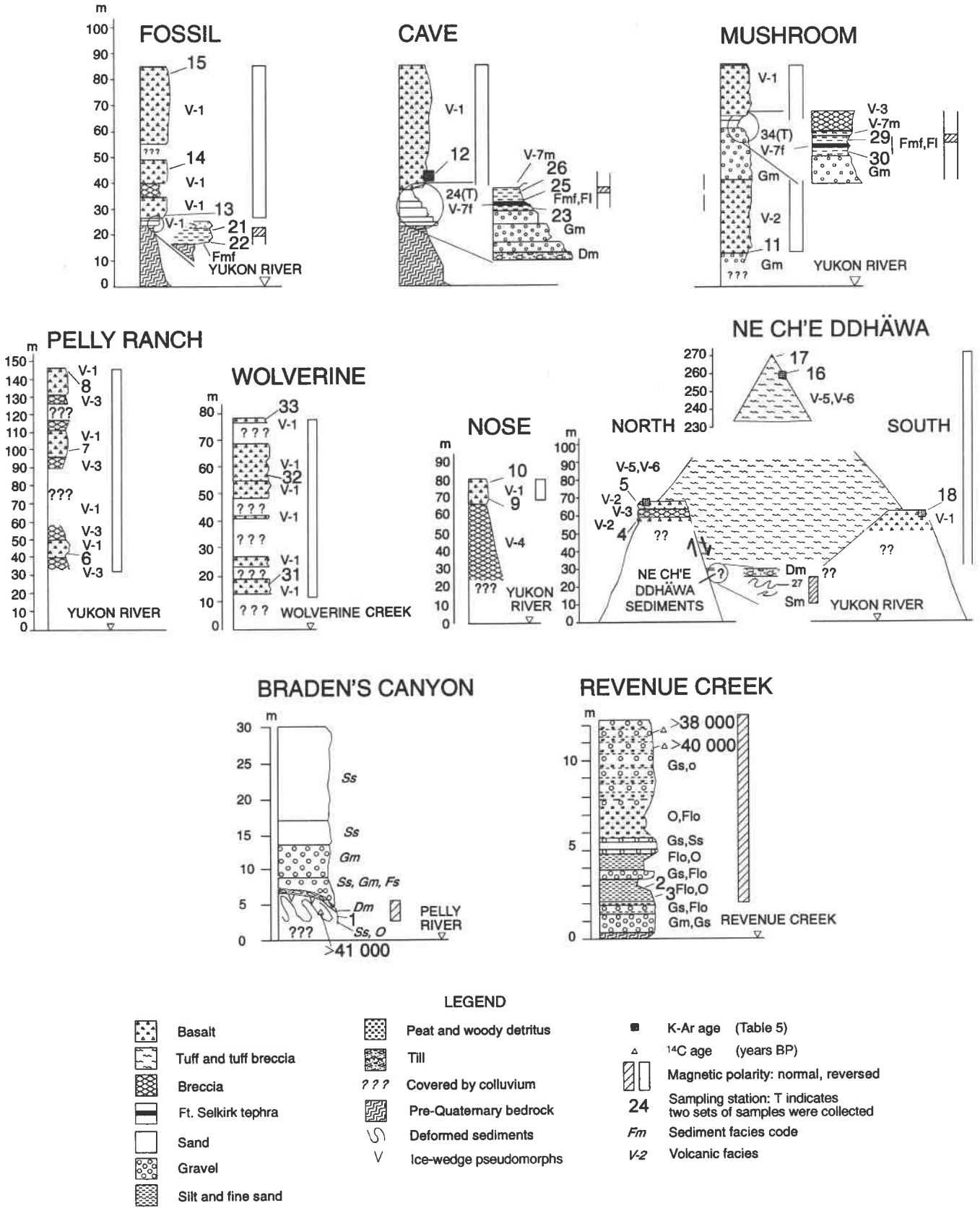
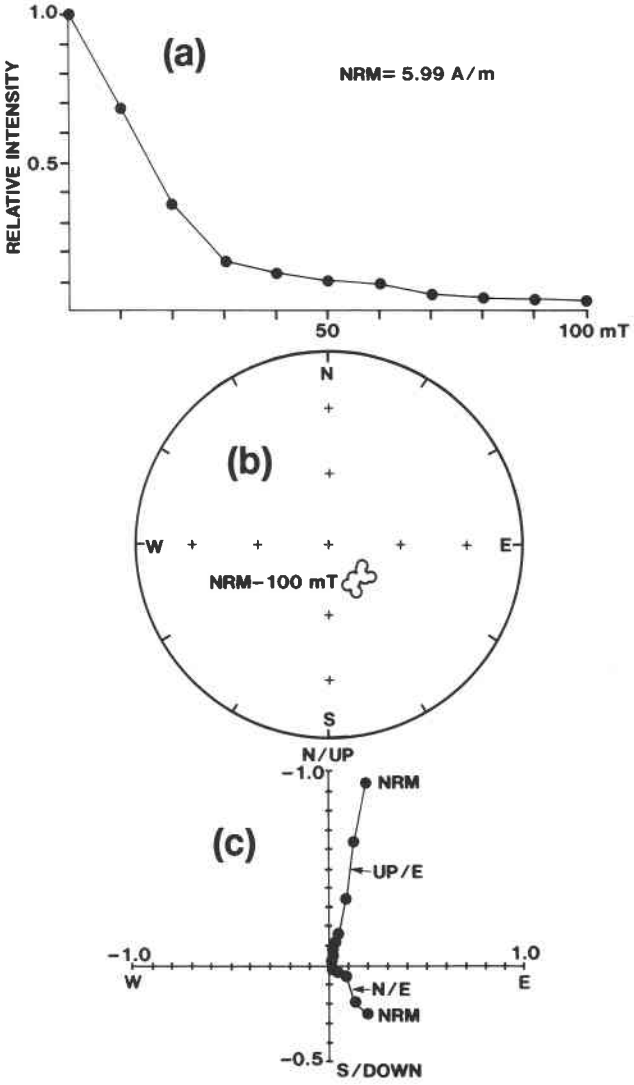


Table 3. Sedimentary facies and environments, Fort Selkirk area.

Facies	Texture	Stratification and sedimentology	Depositional environment
Dm	Stony diamicton, sandy to clay loam matrix	Massive and fissile; has a strong unimodal pebble fabric; stones are faceted and striated and have a regional provenance; underlying sediments are sheared, folded, and thrusted	Glaciogenic: lodgement or melt-out till based upon texture, stratification, and sedimentology (Dreimanis 1989; Ward 1989)
Gm, Gs	Poorly sorted gravel and minor sand (Mushroom section (upper gravel), and Cave section)); well-sorted gravel (Mushroom section (basal gravel)	Massive (Gm) and planar (Gs) Mushroom section (upper gravel) and Cave section: poorly sorted; clasts angular and have a regional provenance; conformably overlies Dm	Glaciofluvial: distal outwash based upon stratification and sedimentology and location of sediments 50 m above Yukon River
		Mushroom section (basal gravel): clasts rounded, imbricated, and consist of resistant local and regional lithologies	Fluvial or distal glaciofluvial; may predate glaciation?
		Revenue Creek: poorly sorted; clasts angular, with lithology restricted to the Revenue Creek drainage basin	Fluvial: braided stream – alluvial fan based upon local clast lithology, angularity, and incised valley environment of the deposits
Sm, Sg, Ss	Medium to coarse sand	Massive (Sm), normally graded (Sg), and planar bedded, rippled or cross-stratified (Ss); the three facies are usually interstratified; beds are lensoidal and intercalated	Fluvial (Braden's Canyon section below Dm, Revenue Creek, and Ne Ch'e Ddhäwa sediments sections): bar, channel, and lacustrine flood-plain deposits; palynological and other paleoenvironmental data indicate deposition during interstade or interglacial (Ward 1989; Mott, written communication, 1990; H. Jetté, written communication, 1995; A.M. Telka, written communication, 1995); glaciofluvial and fluvial (Braden's Canyon section above Dm): McConnell Glaciation outwash and Holocene stream-channel deposits (Ward 1993)
S-o	Sand, peat, logs, woody debris	Planar bedded, cryoturbated	Fluvial and colluvial (Revenue Creek): flood-plain channels and bars and toes of colluvial slopes; deposition under alternating glacial and interstadial climatic regimen (H. Jetté, written communication, 1995; A.M. Telka, written communication, 1995); fluvial (Braden's Canyon section): bar and channel deposits (Ward 1993)
Fm	Inorganic silt and very fine sand	Massive: at Fossil section, texture 85 wt.% within the 4–8 μm size range	Aeolian: loess, based upon texture alone (Swineford and Frye 1945; Péwé 1951); limb bones of *Rangifer* sp. (caribou) and a lagomorph (rabbit or hare) about the size of *Lepus arcticus* (C.R. Harington, personal communication, 1991) found in sediments at Fossil section and teeth from *Lasiopodomys* sp. (primative vole) at Cave section are consistent with an aeolian environment and Early Pleistocene age (C.R. Harington, personal communication, 1993)
Fl	Inorganic silt and very fine sand	Laminated to thinly bedded	Lacustrine: includes resedimented loess; sediments at Cave section contain laminae of degraded moss
Fl-o	Organic and inorganic silt (muck) and very fine sand	Laminated to thinly bedded; sediments cryoturbated	Fluvial: flood-plain and colluvial deposition under alternating glacial and interstadial climatic regimen (H. Jetté, written communication, 1995; A.M. Telka, written communication, 1995)

Notes: Sections refer to Fig. 4.

Fig. 5. Demagnetization of basalt from the upper flow at site 10 (Nose locality). (*a*) Changes in intensity with increasing alternating fields. (*b*) Directions observed at each demagnetization step. (*c*) Orthogonal plots in the horizontal (N/E) and vertical (UP/E) planes. Reversed magnetization referred to the Matuyama.



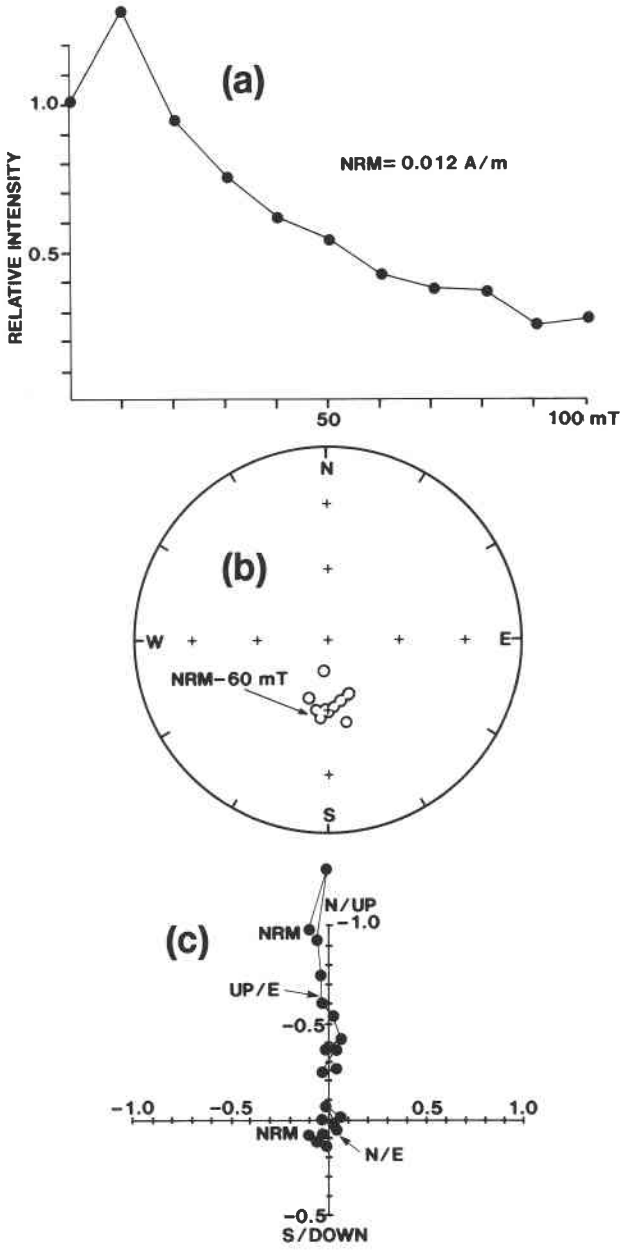
represent glacial, glaciofluvial, fluvial, and eolian environments (Table 3; Fig. 4).

Paleomagnetism

Oriented cores were obtained from basalts. One or two cylindrical specimens were cut from each core. Sediments were sampled by pushing plastic cubes or cylinders into a vertical face that had been previously cleaned by shovel and knife. Orientation was determined by magnetic compass with occasional checks by sun compass. A total of 82 cores and 141 cubes were studied.

Measurements of natural remanent magnetization (NRM) were made on fully automated Schonstedt spinner magnetometers. Stepwise alternating field (AF) demagnetization was carried out using a Schonstedt GSC-5 for fields up to 100 mT, and Sapphire Instrument's SI-4 for fields up to 180 mT.

Fig. 6. Demagnetization of specimen from the Fort Selkirk tephra, site 24 (Cave locality). Brunhes-age overprint on reversed magnetization referred to the Matuyama. See caption to Fig. 5.



Demagnetization of sediment cubes was carried out along three axes successively. Stepwise thermal demagnetization was carried out using Schonstedt TSD furnaces.

In our earlier study, three types of magnetization, namely A, B, and C, were observed (Jackson et al. 1990) (Figs. 5–7). A-type magnetization consisted of a single component with a high remanent coercive force on which small Brunhes-age viscous remanent magnetizations (VRM) were superimposed. The latter could be readily removed in low alternating fields. The former were considered to be accurate recorders of the paleofield. A-type magnetizations occur in basalt, fine-grained loess, and lacustrine sediments. The less reliable, often highly scattered magnetization of B and C type showed poor decay

lines in orthogonal plots (see Jackson et al. 1990). Generally they were found in coarser grained or glacially tectonized sediments. Therefore, when sampling for the present study, the latter were avoided, and only basalts and undeformed fine-grained sediments were selected: most of these magnetizations are of A type. However, a few B-type (10%) and C-type (5%) magnetizations occurred (Fig. 4).

At each site trial specimens were demagnetized in detail (Figs. 5–7). Orthogonal plots showed linear decay to the origin in alternating fields above 10 mT. Five or more additional specimens were then selected from each site and demagnetized in several steps (usually 4) in the range (usually 10–60 mT) in which there was linear decay in trial specimens. This was done to determine whether linear decay to the origin occurred throughout the site. This was generally found to be so (A-type specimens 85%), and the remaining specimens were then cleaned at a field chosen from the early part of the decay line. In B-type specimens (10%) decay lines were less good and they were estimated from end points (see Fig. 6 in Jackson et al. 1990).

The results are listed in Table 4, which includes data of Jackson et al. (1990). Site directions are plotted in Fig. 8. Sediments and lavas have mean directions that are, within errors, 180° apart, indicating that true reversals of the field are recorded. Mean inclinations of sediments and volcanics agree, so the former have no inclination error. The overall mean inclination (bottom of Table 4) departs significantly from that of the geocentric axial dipole field (75.5°). The difference in inclination is small (1.5° outside error). There is a westerly bias (6° outside error) in declinations arising from systematic westerly bias in the lavas. These could have been erupted over a time insufficient to average secular variation. There is no bias in declination of sediments that represent a much wider time range.

Magnetic polarity and orientation of Selkirk Volcanics and sedimentary successions

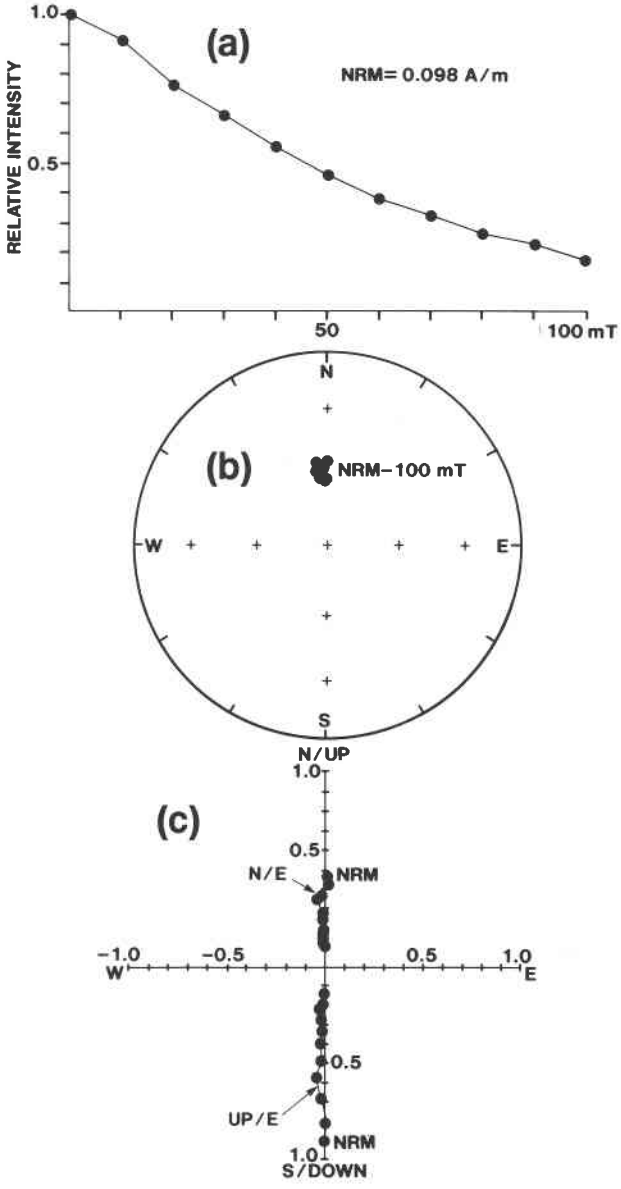
The oldest unit is the gravel (Gm) that underlies the facies V-2 basalts at Mushroom section (Fig. 4). There are no paleomagnetic or radiometric data from this unit. Obvious indications of a glacial genesis are absent. It could predate regional glaciation in central Yukon.

Reversely magnetized V-2 basalt and pillow breccia overlie this gravel at Mushroom section (Fig. 4). It is the oldest component of the Selkirk Volcanics.

The next youngest unit is glacial drift from the older of the two pre-Reid glaciations. The most complete sequence is at the Cave section where till (Dm) is overlain by gravelly outwash (Gm). Only the outwash is present at the Mushroom section. The drift at both sections is too coarse for paleomagnetic analysis.

Normal and reversed nonglacial silt and fine sand, including the Fort Selkirk tephra, overlie the older pre-Reid glaciation drift in the Cave and Mushroom sections. The directions of magnetization of reversed sediments are not significantly different from those of the overlying basalts. Directions in the normally magnetized sediments differ within error limits, by 180° (Table 4), and therefore record a true reversal, not a field excursion. Fort Selkirk tephra is not present within

Fig. 7. Demagnetization of specimen from loess, site 21 (Fossil locality). Normal magnetization referred to the Jaramillo Subchron (see caption to Fig. 5).



the normally and reversely magnetized silt (loess) at the Fossil section. However, the loess occurs in a stratigraphic position similar to that of loessal sediments in the Cave and Mushroom sections. We cannot be sure that they are of the same age because the loess at the Fossil section rests on bedrock, whereas at the Cave and Mushroom sections the silts and fine sands containing the Fort Selkirk tephra are underlain by outwash.

Sands immediately underlying Ne Ch'e Ddhäwa (Fig. 4, station 27) have a stratigraphic position similar to that of the sand and silt beds just described. These bear an interstadial or interglacial pollen assemblage (R.J. Mott, written communication, 1989). They have normal polarity, but the directions of magnetization differ greatly from all other sediments and basalts, apparently due to deformation by glacial overriding (Jackson et al. 1990). A till lying between the sands

Table 4. Paleomagnetic data by sites.

Collecting site	UTM (easting and northing, zone 8V)	Polarity	n	M (A/m)	D (°)	I (°)	k	α_{95} (°)	Assigned age[a]
Braden's Canyon (sediments)									
1. Sands beneath till[b]	406200,6960200	N	5	0.12	325	73	44	6	B
Revenue Creek (sediments)									
2. Upper silts	382300,6913700	N	17	0.05	309	62	61	5	B
3. Lower silts	382300,6913700	N	14	0.03	330	77	96	4	B
Fossil locality (sediments)									
21. Upper loess	374800,6966900	N	8	0.06	356	58	36	9	J
22. Lower loess	374800,6966900	R	19	0.03	160	−69	36	6	mM
Cave and Mushroom localities (sediments)[c]									
26. Upper silts, cave[b]	375100,6966800	N	11	0.05	030	73	74	5	J
29. Upper silts, mushroom	375100,6966200	N	8	0.07	013	79	129	5	J
25. Lower silt above tephra, cave[b]	375100,6966800	R	56	0.04	168	−62	22	4	mM
23. Fine sand below tephra, cave[b]	375100,6966800	R	7	0.02	198	−66	20	14	mM
30. Lower silts, mushroom	376100,6966200	R	15	0.05	207	−69	17	10	mM
24. Tephra, cave	375100,6966800	R	17	0.07	197	−73	44	5	mM
34. Tephra, mushroom	375100,6966200	R	2	0.04	209	−68	86	27	mM
Ne Ch'e Ddhäwa north site (glacial sediments)									
27. Deformed sands and silts	374800,6966900	N	10	0.004	057	63	29	9	J or pJ
Selkirk Volcanics									
17. NCD volcano, tuff breccia[b]	384500,6959500	R	5 (10)	19.0	179	−72	1162	2	lM
16. NCD volcano, pillow basalt[b,d]	384300,6959500	R	23 (23)	10.0	167	−73	452	4	lM
NCD volcano combined		R	28	12.7	168	−72	109	3	lM
15. Fossil locality, upper basalt	375100,6966800	R	7 (14)	2.3	156	−63	207	4	lM
14. Fossil locality, middle basalt	375100,6966800	R	6 (12)	3.4	166	−65	138	6	lM
13. Fossil locality, lower basalt	375100,6966800	R	6 (12)	4.1	144	−65	212	5	lM
12. Cave locality basalt[b]	375100,6966800	R	14 (24)	2.0	172	−72	187	3	lM
5. Fort Selkirk basalt at NCD north[b]	383400,6960300	R	8 (15)	1.0	197	−73	175	4	lM
4. Fort Selkirk basalt at NCD north[b]	383400,6960300	R	8 (13)	0.7	196	−71	269	3	lM
18. Fort Selkirk basalt at NCD south[b]	383600,6958500	R	11 (20)	1.3	166	−71	512	2	lM
10. Nose locality, upper basalt	380600,6963600	R	12 (24)	5.8	139	−70	224	3	lM
9. Nose locality, lower basalt	380600,6963600	R	8 (16)	5.5	166	−70	168	4	lM
8. Pelly Ranch, upper basalt	380900,6968500	R	15 (28)	4.0	120	−72	93	4	lM
7. Pelly Ranch, middle basalt	380900,6968500	R	14 (25)	3.8	167	−73	235	3	lM
6. Pelly Ranch, lower basalt	380900,6968500	R	10 (20)	2.8	170	−70	458	2	lM
33. Wolverine Creek, upper basalt	379200,6954700	R	6 (6)	2.0	158	−66	228	8	lM
32. Wolverine Creek, middle basalt	379300,6954600	R	4 (4)	3.1	178	−66	65	15	lM
31. Wolverine Creek, lower basalt	379500,6954400	R	6 (8)	2.1	184	−74	95	6	lM
11. Mushroom locality, lower basalt	377100,6964400	R	10 (19)	1.7	133	−83	62	6	mM?
Normal sediments	—	N	7[e]	—	357	73	27	12	
Reversed lavas and sediments	—	R	22[e]	—	170	−70	104	3	
Selkirk Volcanics combined	—	R	17[e]	—	165	−71	126	3	
All sediments combined[f]	—	NR	12[e]	—	001	71	39	7	
Sediments and lavas combined[g]	—	NR	29[e]	—	352	71	64	3	

Notes: Site numbers have no stratigraphic significance. n, number of oriented samples from sediments or cores from lavas (number of specimens from lavas given in parentheses); M, intensity of NRM; D and I, declination and inclination of mean direction; k, Fisher's estimate of precision; α_{95}, radius of circle of confidence ($P = 0.05$); NCD, Ne Ch'e Ddhäwa.

[a]B, Brunhes; lM, late Matuyama; J, Jaramillo; mM, middle Matuyama; pJ, any of the Pleistocene magnetically normal intervals predating Jaramillo.

[b]Data of Jackson et al. (1990).

[c]Normally magnetized sites making up the top one third of Cave locality section at both localities are referred to as upper silts, and reversely magnetized silts making up the lower two thirds are referred to as lower silts. The Fort Selkirk tephra lies within the lower silts.

[d]Mean of 4 sites (23 cores) spaced through 10 m of the same volcanic unit (Jackson et al. 1990).

[e]Sites each given unit weight.

[f]Mean of entries 1, 2, 3, 21, 22, 23, 24 (+34), 25, 26, 27, 29, and 30 (12 sites), irrespective of sign.

[g]Mean, irrespective of sign, of all sites included in previous entry and 17 volcanic units distributed through last 1.8 Ma. The lower basalt at the Mushroom locality is not included because of its uncertain age. Paleopole calculated relative to the centre of the collection area (62.8°N, 137.4°W) is 82°N, 76°E (semi-axes of the 95% ellipse of confidence about the paleopole ($d_m = 6°$, $d_p = 5°$). Inclination of the geocentric axial dipole is 75.5°.

and facies V-6 tuff (base of Ne Ch'e Ddhäwa; Fig. 3*d*) was unsuitable for sampling. Because of this deformation, the sands cannot be confidently correlated with other sediments that are in contact with the Selkirk Volcanics. All that can be concluded is that they were laid down during an interval of normal polarity within the Pleistocene prior to the eruption of Ne Ch'e Ddhäwa.

Basalt flows overlying the sediments at Fossil, Cave, and Mushroom sections and basalt or hyaloclastite constituting the entire sections at Wolverine, Ne Ch'e Ddhäwa, Nose, and Pelly Ranch (Fig. 4) are all magnetically reversed. The close agreements of their directions of magnetization (Fisher's estimate of precision $k = 126$) in these rocks, as noted above, indicate that they were erupted over an interval of perhaps a few centuries. This corroborates the conclusions of Francis and Ludden (1990) that, although the flows filling the valleys of the Yukon and Pelly rivers and Wolverine Creek issued from several vents, they constitute a common eruptive sequence and therefore are likely to be of similar age.

Sediments investigated at Revenue Creek and Braden's Canyon (Fig. 4) are normally magnetized and occur in valleys that were cut or deepened sometime after the pre-Reid glaciations.

Correlations of Selkirk Volcanics with the magnetic polarity time scale and age of the pre-Reid glaciations

Radiometric ages on the Selkirk Volcanics including interstratified sediments come from two sources: whole-rock K–Ar ages on V-1 basalts, and fission-track ages on the Fort Selkirk Tephra (Fig. 4; Table 5). The chronology we propose for the sediments, volcanic rocks, and glacial stratigraphic units with respect to the geopolarity time scale is summarized in Fig. 9. We recognize an early Matuyama interval (pre-Olduvai Subchron), a middle Matuyama interval (between Olduvai and Jaramillo subchrons), and a late Matuyama interval (post-Jaramillo Subchron). We begin by assuming that the normal interval recorded in the nonglacial sediments lying within the Selkirk Volcanics is the Jaramillo Subchron (1.07–0.99 Ma; Shackleton et al. 1990) rather than the Cobb Mountain Subchron (Mankinen et al. 1978), which was very brief (likely only several thousand years). The measurements of the latter's age range from between 1.201 and 1.211 Ma (Cande and Kent 1995) to 1.19 Ma (Shackleton et al. 1990). The possibility that the Cobb Mountain Subchron is represented is considered later.

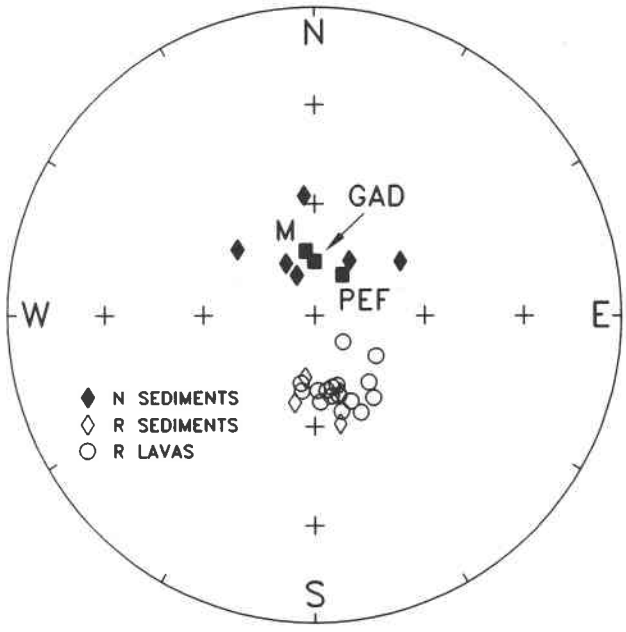
Age of the older pre-Reid glaciation

This glaciation occurred after the eruption of the lowest basalt at the Mushroom section (1.60 ± 0.19 Ma) and prior to the eruption of the Fort Selkirk tephra, that is, it occurred within the middle Matuyama interval. The range of the eight radiometric ages of the Fort Selkirk tephra (1.54–0.84 Ma; Table 5) is in excess of the individual errors, indicating their poor accuracy. We suggest that the youngest of the middle Matuyama ages, 1.19 ± 0.11 Ma, should be regarded as a minimum estimate of the age of the Fort Selkirk tephra and the older pre-Reid glaciation.

Age of the younger pre-Reid glaciation

The magnetically reversed valley-filling basalts (Fig. 4: Cave,

Fig. 8. Mean direction of magnetizations at collecting sites. Solid symbols represent downward inclinations, and open symbols upward inclinations. Beds are flat lying, and perimeter is the horizontal at the collecting sites. GAD, geocentric axial dipole field; PEF, present Earth's field; M, mean of all sites, irrespective of sign.



Fossil, Mushroom, Pelly Ranch, and Wolverine sections) have been shown to be coeval with the Ne Ch'e Ddhäwa hyaloclastite complex, which erupted during the younger pre-Reid glaciation. The underlying normally magnetized silt beds at the Fossil, Cave, and Mushroom sections constrain the age of these basalts to the late Matuyama. Thus, the younger pre-Reid glaciation occurred within the late Matuyama (Fig. 9). This contrasts with four earlier K–Ar ages (1.47–1.08 Ma; Table 5) obtained from flows within this valley-filling complex (Fig. 4, Cave section, station 12). In an attempt to resolve this conflict, four additional ages (3.92–1.28 Ma) were determined on V-1 basalts (Table 5). We argue that these ages are erroneous for the following reasons:

(1) The >2 Ma spread in the ages is inconsistent with the paleomagnetic data (of very high precision, $k = 126$) and the petrological evidence, which indicate that the Selkirk Volcanics were erupted over hundreds of years or less, not millions of years. Ages over 2 Ma would place the subglacial eruptions within the Pliocene, earlier than any known regional glaciation in Yukon.

(2) Mantle-derived xenoliths and xenocrysts (Sinclair et al. 1978) are abundant throughout the Selkirk Volcanics. Elsewhere, the presence of these in young, low-K basalts has been associated with significant excess mantle-derived Ar, causing anomalously old age determinations (Dalrymple and Moore 1968).

(3) All analyses yield high proportions of atmospheric Ar (74–89%). Fractionation of trapped atmospheric Ar may occur during high-temperature bake-out of the samples prior to fusion, leading to an erroneous correction for the atmospheric component and an anomalously old age determination. Young, low-K basalts with high atmospheric Ar are prone to

Table 5. Radiometric ages on basalt and tephra.

(A) K–Ar age determination (this study).[a]

Paleomagnetism sample site	Field No.	Age (Ma BP)	(wt.%)[b]	Radiogenic Ar[b] (cm^3/g)	Atmospheric Ar (%)
12. Cave	010789R1	1.276±0.34	1.03±0.831	0.5106×10^{-7}	74.0
5. Ne Ch'e Ddhäwa	290689R2	2.362±0.86	1.15±0.557	1.057×10^{-7}	89.0
16. Ne Ch'e Ddhäwa	300689R2	3.921±0.11	1.22±0.936	1.857×10^{-7}	79.0
18. Ne Ch'e Ddhäwa	260689R5	2.078±0.055	1.44±0.651	1.166×10^{-7}	83.0

(B) Previously published K–Ar ages on basalt.

Paleomagnetism sample site	Age (Ma)	Source
11. Mushroom	1.60±0.08	Westgate 1989
12. Cave	1.35±0.08, 1.35±0.11, 1.47±0.11	Westgate 1989
12. Cave	1.08±0.05	Naeser et al. 1982

(C) Previously published fission-track ages on Fort Selkirk tephra.

Sample	Age (Ma)	Method	Source
24. Tephra (glass shards)	1.19±0.11, 1.01±0.11, 1.43±0.26, 1.54±0.27	Isothermal plateau	Westgate 1989
24. Tephra (glass shards)	0.84±0.13, 0.86±0.18	Standard fission-track	Naeser et al. 1982
24. Tephra (zircon)	0.94±0.40	Standard fission-track	Naeser et al. 1982

[a]Determined by the Geological Survey of Canada.
[b]Error values are 1σ.

this effect (Baksi 1973; Souther et al. 1984). Consequently, K–Ar ages on basalts of the Selkirk Volcanics are regarded as unreliable and only of value as estimates of maximum age.

Age of sediments at Braden's Canyon and Revenue Creek

The normal polarity of these sediments is interpreted as indicating a Brunhes age. This is based upon their settings as infills in valleys cut into terrain glaciated during the younger (late Matuyama) pre-Reid glaciation. Furthermore, the Revenue Creek sediments contain a rich fossil fauna interpreted as Late Wisconsinan (Harington 1973, 1977, 1989).

Discussion

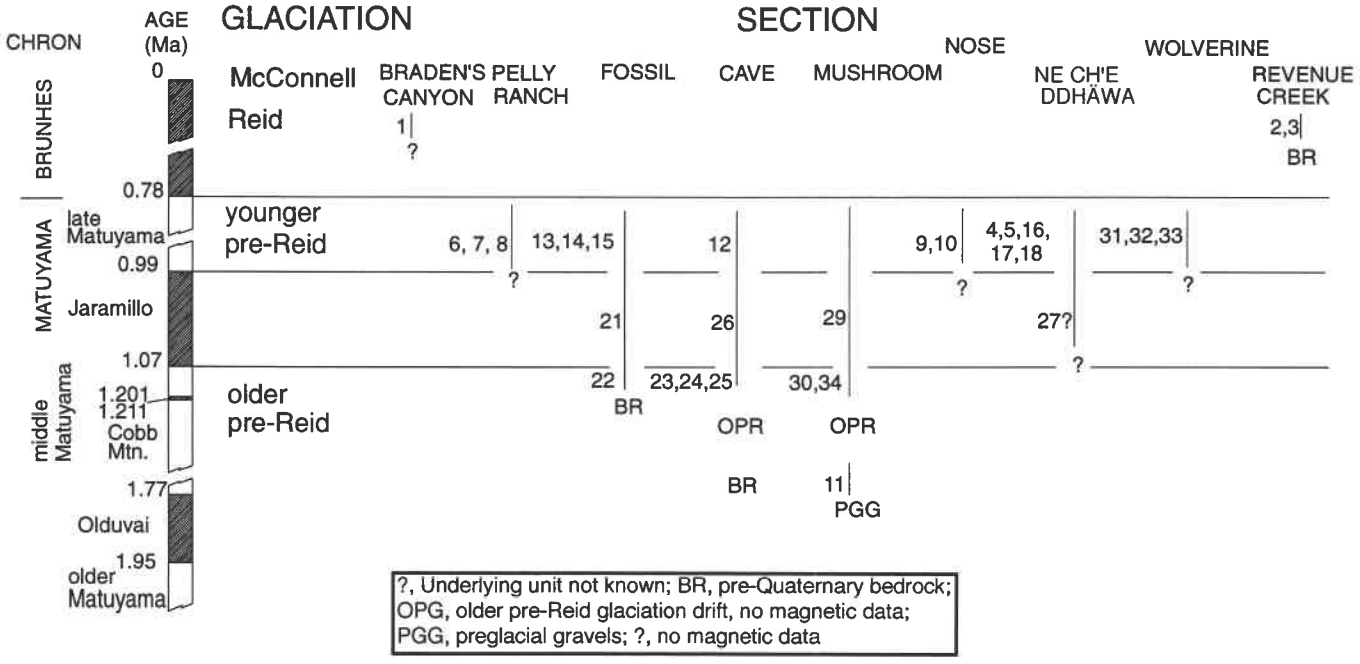
So far, the normally magnetized nonglacial sediment interstratified within the Selkirk Volcanics has been assumed to date from the Jaramillo Subchron. Alternatively, it could date from the Cobb Mountain Subchron. If so, then the older pre-Reid glaciation would remain within the middle Matuyama and the base of the Cobb Mountain Subchron would provide a minimum age for it, independent of the Fort Selkirk tephra. The age of the younger pre-Reid glaciation would be less well constrained. It could date from the middle Matuyama between the end of the Cobb Mountain and beginning of the Jaramillo subchrons (ca. 1.211 and 1.07 Ma, respectively), or from the late Matuyama between the end of Jaramillo and the Matuyama–Brunhes boundary (0.99 and 0.78 Ma, respectively; Ruddiman et al. 1989). However, the overall effect on the chronology of glaciations would be minimal.

Summary

The Selkirk Volcanics were in part subglacially erupted as hyaloclastite tuff and pillow tuff breccia. The paleomagnetism of these volcanic rocks and interstratified sediments was investigated to resolve the geochronology of glaciations. Most of the sediments and basalts sampled accurately record the paleofield. In the Fort Selkirk area, sediments and lavas have mean directions that are, within error limits, 180° apart, indicating that true reversals of the field are recorded. Mean inclinations of sediments and volcanics agree, the former being unaffected by inclination error. There is a very small westerly bias in declinations (352°). This bias is interpreted as the result of eruption of valley-filling lava and hyaloclastite over a very short span of time, which was insufficient to fully average the secular variation.

K–Ar ages determined on this magnetically homogenous sequence have proven to be erroneously old due to excess Ar. All must be regarded as maximum values. Conversely, three out of five isothermal plateau method fission-track ages on the Fort Selkirk tephra are compatible with the magnetostratigraphy. Assuming that the normal magnetic interval recorded within interstratified nonglacial sediments represents the Jaramillo Subchron, the younger pre-Reid glaciation and the main eruptive episode of the Selkirk Volcanics occurred during the late (post-Jaramillo) Matuyama. The older pre-Reid glaciation occurred during the middle Matuyama after ca. 1.6 Ma and prior to the eruption of the Fort Selkirk tephra. If the normal period recorded in these sediments is the Cobb Mountain Subchron, then the younger pre-Reid

Fig. 9. Master correlation plot of sections detailed in Fig. 4 and the magnetic polarity time scale. The correlation scheme assumes sediments at sample locations 21, 26, and 29 were deposited during the Jaramillo Subchron rather than during the briefer Cobb Mountain Subchron.



glaciation could have occurred as early as immediately prior to the Jaramillo at the end of the middle Matuyama or within the younger Matuyama.

Sediments investigated at Revenue Creek and Braden's Canyon are assigned to the Brunhes. They occur in valleys that were cut or deepened sometime after the pre-Reid glaciations. The Revenue Creek sediments contain a vertebrate assemblage that is considered Late Wisconsinan.

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